

# Thermal insulation (RP)

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

|   | QUESTION                                                                                  | ANSWER                                                                                       | ✓                        |
|---|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------|
| 1 | Name the equipment needed to compare thermal insulators.                                  | A beaker of hot water, insulating materials, a thermometer and a stopwatch                   | <input type="checkbox"/> |
| 2 | Comparing thermal insulators: name the independent variable.                              | The type of insulating material                                                              | <input type="checkbox"/> |
| 3 | Comparing thermal insulators: name the dependent variable.                                | The temperature fall of the water in a set time                                              | <input type="checkbox"/> |
| 4 | Comparing thermal insulators: name three control variables.                               | Any three from: volume of water, starting temperature, thickness of insulation, cooling time | <input type="checkbox"/> |
| 5 | Name the piece of equipment used to measure time.                                         | A stopwatch                                                                                  | <input type="checkbox"/> |
| 6 | Comparing thermal insulators: explain why each beaker must start at the same temperature. | Hotter water cools faster, so the comparison would be unfair.                                | <input type="checkbox"/> |